Complex Permittivity Effects in ESP32 Wi-Fi Channel State Information

Advanced Topics in Computer Networks, Griffith University

Nathan Burg
Griffith University

Abstract

Commodity Wi-Fi Channel State Information (CSI) has emerged as a low-cost alternative for non-contact measurement and environmental profiling. Mapping minute variations in complex permittivity of a medium to high-dimensional RF features remains a profound challenge due to hardware instability and low signal-to-noise ratios. This paper formulates and executes a water salinity concentration sensing experiment to expose a systematic methodological confound prevalent in contemporary CSI sensing literature, called the "session fingerprint." We demonstrate that statistical models evaluated on within-session splits—a broadly utilised practice in machine learning—will invariably exploit transient thermal trajectories, hardware gain states, and microscopic geometric variations to achieve inflated or physically impossible accuracy metrics. To suppress these false positives, we propose a differential sensing architecture using a dual-node ESP32-S3 framework. This architecture successfully recovers a coherent salinity gradient in water using cross-session Support Vector Regression (SVR), achieving Mean Absolute Error (MAE) of 1.35%. Our analysis also covers a considerable geometric confound of poorly-calibrated differential sensing arrangements, where parasitic multipath scattering from the target volume couples with the unobstructed reference link, inducing signal erosion in differential feature space. This effect compresses low-concentration features near the quantisation floor of the hardware, reducing overall precision of the model. We conclude that cross-session evaluation is a mandatory validation threshold for wireless material sensing, and outline critical spatial constraints required for future architectures.

1 Introduction

The interaction of electromagnetic radiation with matter is governed primarily by the relationship between wavelength and the physical scale of the medium. At optical frequencies, wavelengths are comparable to the size of atmospheric particulates—such as water droplets, dust, and smoke—which leads to visible attenuation or diffusion called Mie scattering. Long wavelength radio waves below the microwave range pass through particulate media without meaningful scattering, but are too long relative to discrete targets for specular reflection in the general case. Microwave radiation is a practical equilibrium for communication and remote sensing, with wavelengths long enough to penetrate atmospheric particulates and thin dielectric surfaces such as walls that would otherwise attenuate higher-frequency signals. For this reason, microwave frequencies are the foundation of all practical radar systems and terrestrial wireless networks.

The 2.4 GHz ISM band has a wavelength of 12.5 cm, which interacts measurably with dielectric boundaries in the propagation path—boundaries of different permittivity produce reflection, refraction and absorption effects that are encoded in the received signal. The 802.11n standard exposes this per-subcarrier channel response as Channel State Information (CSI), providing amplitude and phase measurements across 54 OFDM subcarriers spanning the 20 MHz channel. Perturbation in the propagation paths of the waves, then, can be directly observed within CSI data.

The passive sensing potential of Wi-Fi CSI has been widely recognised, with the existing literature focusing primarily on spatial localisation [1] and motion detection [2]. These applications exploit the multipath structure of the channel, where the phase and amplitude across subcarriers encode a coarse, high-dimensional representation of the geometry of reflectors in the environment. Changes to the geometry, such as a person or object in motion, produce detectable perturbations in the channel response. This class of application treats the Wi-Fi link as a coarse radar, leveraging reflected energy from targets in the Fresnel zone without any modification to standard hardware or protocols. More complex implementations rely on high-dimensional multipath data for improved coverage and resolution.

Less explored is the effect of changes in the transmission properties of the medium itself, rather than its geometry [3]. When a dielectric material occupies the propagation path, it modifies the channel response through absorption and phase shift determined by its complex permittivity. Water is a particularly strong target for CSI sensing at 2.4 GHz. Its real relative permittivity $\varepsilon' \approx 78$ is far above that of air, producing significant reflection, refraction and attenuation along any link passing through or near a water body. Dissolving ionic salts raises the conductivity σ substantially, from approximately 5×10^{-3} S/m for deionised water to over 14 S/m for a 15% NaCl solution [4], which changes both the attenuation and the phase velocity of electromagnetic waves at 2.4 GHz. In principle, these changes are large enough to be detectable within subcarrier-level CSI.

A primary challenge in commodity CSI sensing is the geometric coupling problem, which typically necessitates extensive calibration or differential multi-node architectures to reject background noise [5]. In controlled environments, this technique can be directly leveraged for non-contact salinity profiling through plastic or glass conduits. By eliminating the need for physical probes, which are highly susceptible to corrosion and biofouling in saline environments, this framework offers a low-cost, low-maintenance alternative for industrial fluid monitoring.

This paper makes three contributions. First, we demonstrate that the salt concentration of a water sample can be resolved from differential ESP32 CSI using interpretable, physically-motivated features and classical machine learning. Second, we show that Support Vector Regression (SVR) is the epistemically appropriate model for this task, in favour of classification models, as salinity is a continuous physical property and discrete classification discards ordinal structure and produces overconfident class boundaries. Unlike spatial sensing, where temporal effects are the primary correlation, transmission through a static dielectric target is highly sensitive to noise and temporal drift. Third, and most significantly, we identify a methodological confound that is systematically present in a wide body of CSI-sensing literature. We demonstrate that within-session evaluation (train-test split within one session) produces near-perfect accuracy that collapses entirely on a held-out session. The confound is a session fingerprint, where hardware thermal drift and position-specific multipath structure is stable enough within a session to be exploited by any sufficiently expressive model, but does not transfer across sessions. The accuracy figure reported under within-session evaluation therefore reflects this fingerprint, not the target physical property, and does not constitute evidence of sensing capability.

2 Theory and Prior Work

Detecting the transmission effect of dielectric materials requires three primary modes of understanding. First being the physics of microwave frequencies, and how they are affected by dielectric and conductive media. Second being the structure of high-dimensional CSI data, what signal it measures, and what feature space is favoured. Third being the application of machine learning on this data, and what failure modes are introduced by high-dimensional, low-signal-to-noise ratio data like CSI.

2.1 Complex Permittivity

The interactive relationship described earlier, between the wavelength and physical scale of a medium, can be defined as the complex permittivity of a material.

$$\varepsilon^* = \varepsilon' - j\varepsilon'' \tag{1}$$

where ε' is the real (dispersive) part, describing how strongly the material polarises under an applied field and thereby governing the phase velocity of a wave passing through it, and ε'' is the imaginary (absorptive) part, capturing energy dissipated per cycle [6, 7]. The phase velocity inside the material is

$$v_p = \frac{c}{\sqrt{\varepsilon'}} \tag{2}$$

so a higher real permittivity slows the wave and increases the phase accumulated over a given path length.

Water is a polar molecule whose dipoles rotate under an applied electric field. At low frequencies this rotation is nearly instantaneous and ε' is large (∼80). At 2.4 GHz, the field reverses 2.4 billion times per second, and molecular rotation begins to lag behind the field. The resulting molecular friction dissipates energy directly as heat and produces a large imaginary component $\varepsilon'' \approx 12$ at room temperature—a mechanism called the Debye relaxation [7, 8]. This is the physical principle of microwave heating, where high energy microwave radiation produces heat through molecular friction in water.

The consequence for propagation is captured by the skin depth δ, the distance over which wave amplitude decays to $1/e$ of its initial value [6]:

$$\delta = \frac{1}{\omega}\sqrt{\frac{2}{\mu|\varepsilon^*|}}\left(\sqrt{1+\left(\frac{\varepsilon''}{\varepsilon'}\right)^2}-1\right)^{-1/2} \tag{3}$$

For pure water at 2.4 GHz, accounting for both the relaxation loss $\varepsilon'' \approx 12$ and the real part $\varepsilon' \approx 77$, this yields $\delta \approx 2.9$ cm, consistent with empirical measurements [9]. A 15 cm container therefore attenuates the transmitted amplitude by a factor of $e^{-15/2.9} \approx 0.006$, which is effectively total extinction.

Dissolving NaCl introduces free Na^+ and Cl^- ions that increase ionic conductivity σ, adding a further loss term $\sigma/\omega\varepsilon_0$ to ε'' and reducing the skin depth further. Salt also disrupts the hydrogen bonding network of water, modifying the Debye relaxation parameters and reducing ε' from ∼77 (pure water) to ∼62 at 15% NaCl [4, 10, 11]. Both effects alter the complex permittivity monotonically with concentration. The practical consequence is that the surface reflection coefficient at the boundary

$$\Gamma = \frac{\sqrt{\varepsilon^*}-1}{\sqrt{\varepsilon^*}+1}, \tag{4}$$

changes with salinity, as does the edge diffraction pattern around the container. Because the direct transmission path is extinguished for any practically-sized water container at 2.4 GHz, these surface interaction effects constitute the primary propagation mechanism of Wi-Fi, and these salinity-dependent changes in the multipath structure may be separable within the CSI subcarrier response [3].

2.2 Channel State Information Sensing

2.2.1 Definition

CSI sensing is the application of subcarrier amplitude and phase for detecting the interaction of objects within a high-dimensional multipath environment [1]. In an 802.11n OFDM system, the channel response on subcarrier k is

$$H_k = \alpha_k \, e^{j\phi_k} \tag{5}$$

where α_k is the amplitude attenuation and ϕ_k is the phase response. Modern implementations expose H_k as in-phase and quadrature (I/Q) components, from which amplitude and phase can be recovered [12].

Raw CSI phase is contaminated by Carrier Frequency Offset (CFO) and Symbol Timing Offset (STO), which impose a linear phase ramp across subcarriers

$$\hat{\phi_k} = \phi_k + mk + b + \text{noise} \tag{6}$$

The slope m and intercept b are estimated by fitting a line to the unwrapped phase profile across active subcarriers and then subtracted. The residual $\phi_k - mk - b$ represents the physical channel respsonse free of hardware-induced offsets [1, 13]. In practice, these offsets are not static—the oscillator, which governs the carrier frequency of the transmitter, drifts with device temperature, meaning m and b drift continuously with temperature over time. This drift is internally consistent within a recording session, but becomes separable between sessions, and constitutes one component of the session fingerprint discussed in Section 2.2.2.

2.2.2 Geometric Sensitivity and Replicability

The received signal is a superposition of contributions from all propagation paths

$$H_k = \sum_i \alpha_i e^{j\phi_{k,i}} \tag{7}$$

where each path i contributes amplitude α_i and subcarrier-dependent phase $\phi_{k,i} = 2\pi k \Delta f \tau_i$, with Δf as the subcarrier spacing and τ_i the path delay. Because the wavelength at 2.4 GHz is 12.5 cm, a displacement of any reflector by $\lambda/4 \approx 3\,\text{cm}$ changes the phase contribution of that path by $\pi/2$ radians, which is sufficient to invert the interference relationship on that subcarrier. Sub-centimetre positional changes between recording sessions therefore produce CSI perturbations that are strongly separable in feature space.

This geometric sensitivity has a direct consequence for the replicability and generalisability of CSI sensing results. A model trained on data from one physical environment encodes the specific multipath geometry of that site as implicit discriminative features. Unlike visual features in image data, which retain semantic meaning across environments, CSI features are entangled with the recording geometry in a way that is inseparable from the target signal, which makes cross-deployment transfer exceedingly difficult [5, 14]. Results obtained in one laboratory cannot be assumed to replicate in another, and performance degradation under environmental change is difficult to predict or bound.

The combination of thermal drift and geometric sensitivity produces a characteristic session fingerprint, which is a highly separable feature of the recording session itself, regardless of the sensing target. Any sufficiently expressive model—particularly deep learning models—that is trained and evaluated within a single session will preferentially exploit this fingerprint over the weaker physical signal, because it is high signal-to-noise and requires no physical understanding of the target to classify [15].

2.2.3 Differential Sensing

Differential sensing is a technique used to isolate a desired signal in a high-noise environment. To mitigate the effect of temporal drift and session fingerprinting, we can use a control signal on an identical device under different conditions. For each subcarrier k, the differential channel is

$$\Delta H_k = \frac{H_{\text{obs},k}}{H_{\text{ref},k}} \tag{8}$$

where $H_{\text{obs},k}$ is the channel response on the obstructed link and $H_{\text{ref},k}$ is the simultaneously recorded response on an unobstructed reference link. Complex division cancels the transmitter's common-mode response, including transmit power, antenna pattern, and shared CFO, leaving a per-subcarrier ratio whose amplitude reflects the relative attenuation introduced by the target, and whose phase reflects the differential phase shift [16]. The reference link therefore plays the role of a real-time baseline, suppressing environmental changes that affect both links equally, such as temperature-driven oscillator drift and slow multipath variations from air movement or thermal expansion of the environment.

This is not a complete solution to the session fingerprinting problem. Each device maintains an independent oscillator with its own thermal trajectory, and while identical hardware exhibits similar drift characteristics, the residual inter-device phase difference is non-zero and varies between sessions [17]. The differential channel suppresses the shared temporal component of the fingerprint, but the site-specific multipath unique to each link does not cancel in the ratio. Cross-session evaluation therefore remains the necessary condition for validating that a model has generalised to the physical signal, rather than residual session artefacts.

2.3 Classification, Regression, and Deep Learning

2.3.1 Standard Modelling

Support Vector Machines with an RBF kernel are a natural baseline for classification of CSI data. The kernel implicitly embeds the feature space into high dimensions, and the max-margin criterion generalises better than nearest-neighbour methods when cross-session drift shifts the feature distribution [18]. For concentration problems, Support Vector Regression (SVR) is the appropriate model. In this case, water salinity is a continuous, ordinal property, and although the transmission effects within the propagation path of Wi-Fi radiation may be high dimensional and non-linear, discarding this ordinal property introduces artificial hard boundaries that classifier models may derive undesirable signals from. SVR predicts a real-valued output and its performance is measured as mean absolute error (MAE) in physical units, which is easily interpretable [19].

2.3.2 Deep Learning and its Misapplication

Convolutional and recurrent networks applied to raw or lightly processed CSI are increasingly common in the sensing literature [1]. The classification strength of deep learning and information density of CSI superficially appear conducive to this. The architecture can learn discriminative features without manual feature engineering, and accuracy figures >90% are frequently reported in held-out test data [20–22].

The critical failure mode, which we encounter through our experimentation and demonstrate empirically in Section 4, is that these models are evaluated without ablating the confounding signals encoded in the CSI. Every recording session introduces what we have called a session fingerprint:

1. Thermal drift of the receiver device, which affects the radio oscillator and perturbs the I/Q rotation as the device changes temperature.
2. Position-specific multipath structure from the surrounding environment, including thermal expansion and permittivity, which may consistent within a recording, but not between them—we observe that these artefacts are separable with deep learning.
3. The transmitter characteristics of the router, where effects like Automatic Gain Control (AGC) and switch between sessions, resulting in an obvious categorical fingerprint.

When training and test data are drawn from different time windows of the same session, which is common in non-temporal machine learning applications like image classification, a sufficiently expressive model can exploit session-internal structure of timeseries data to consistently achieve high accuracy without ever resolving the intended signal [15]. This differs from standard overfitting—rather, the model generalises perfectly to the test split precisely because the session fingerprint is consistent across that split.

The necessary ablation is simply to evaluate the model on a held-out session. If within-session accuracy is high while cross-session accuracy is near chance, the model has learned the session fingerprint rather than the desired signal regardless of how the split was constructed, or to what ratio. In the general case, if a result cannot survive this test, the physical claim is unsupported.

This ablation is completely absent from a number of published works on CSI material and environment sensing [20–22]. High classification accuracy is reported on commodity Wi-Fi CSI using single session recordings, evaluating within-session only, and interpreting the high accuracy as support of an underlying physical hypothesis. Without a cross-session evaluation, no such conclusion can be drawn.

3 Methodology

3.1 Proposal

This research aims to answer whether a container of water in the direct propagation path of commodity Wi-Fi devices, at varying saline concentrations, is differentiable within CSI subcarrier data.

Fresh water at 2.4 GHz has a skin depth of $\delta \approx 2.9$ cm, accounting for both Debye relaxation loss ($\varepsilon \approx 12$) and ionic conductivity. A 15 cm container therefore attenuates the transmitted amplitude by a factor of $e^{-15/2.9} \approx 0.006$, which effectively eliminates the direct path signal. Increasing salinity reduces the skin depth further. At 3.5% NaCl, $\delta \approx 1.37$ cm, giving attenuation of $e^{-15/1.37} \approx 10^{-21}$. The direct transmission path is negligible at all tested concentrations.

This effect is a monotonic gradient on complex permittivity ε^* of the water, where the real permittivity ε' is reduced from about 80 for pure water, to about 72 for 3.5% NaCl solution, and imaginary loss ε'' is increased through the growing ionic conductivity of the solution. Free Na^+ and Cl^- ions contribute a conductivity term $\sigma/\varepsilon_0\omega$ that becomes dominant in microwave frequencies above approximately 1% concentration.

Because the sensing signal arises from salinity-dependent changes in surface reflection and edge diffraction rather than direct transmission, it is distributed across the multipath structure of the channel rather than encoded in total received power. A scalar measurement such as RSSI would be insensitive to these changes, as the reflected and diffracted components are not present in total power. The per-subcarrier amplitude and phase response of CSI resolves these components across frequency, providing the high-dimensional representation necessary to discriminate between concentrations whose surface reflection coefficients differ by small but consistent amounts across subcarriers. We can calculate the direct occlusion from the object using the first Fresnel zone between transmitter and receiver

$$r_F = \sqrt{\frac{\lambda d_1 d_2}{d_1 + d_2}} \tag{9}$$

where $\lambda \approx 12.5$ cm at 2.4 GHz and d_1, d_2 are the transmitter-to-object and object-to-receiver distances respectively. Even though the object obstructs direct line-of-sight, wave diffraction and scattering events permit some signal with a distinct signature.

As this signal is expected to be far weaker than other CSI sensing activities, such as human movement, we must ensure control of environmental variables. For small collection windows, environmental reflection and scattering will have a stronger effect than the secondary multipath signal of the water. Before formulating an experimental setup, there are first-order variables we must control for:

1. Transmitter, receiver and container position must be constant between recordings. In the case of our experiment, the position and orientation of the container have a stronger effect on CSI data than the permittivity of the medium.
2. External interference must be minimised during recordings. Human traffic, external Wi-Fi interference, and other interfering environmental signatures must not be present in the collected data.
3. Recording sessions must allow sufficient hardware warm-up time before data collection begins. Oscillator drift during thermal stabilisation produces non-stationary phase offsets that cannot be removed with standard signal processing techniques. This would contaminate features with a time-varying artefact uncorrelated with the sensing target.
4. Data must be collected across multiple independent sessions to enable cross-session evaluation. Within-session train-test splits are insufficient to validate physical sensing capability.
5. Samples must be externally timed, as commodity hardware suffers from unreliable internal clocks, and inconsistent sampling rates, which strongly contaminates sampling data.

Insufficiently rigorous environmental control leads to unpredictable and inconclusive model performance on sampling data, on account of its high dimensionality. Unlike motion detection, the signature of media permittivity in CSI is tightly coupled with the geometry, where the shape and cadence of the noise floor is itself the signal.

3.2 Experimental Setup

The experiment relies on CSI data. To collect this, we use low-cost commodity hardware—approximately analogue to smartphone Wi-Fi—in the form of the Espressif ESP32-S3 microcontroller. To extract the data, we use the 'esp-csi' open-source program provided by Espressif, and read the serial output in real-time using a custom Python program. The ESP32 program was configured in 'router-recv' mode, where a continuous stream of ping packets is delivered to a standard Wi-Fi router, and CSI data is gathered from the response. This setup is common in dielectric sensing experiments. The router used was a Netgear Nighthawk AC750, configured to the 20 MHz band on channel 6 at 2.4 GHz. An improvement to the data collection pipeline was made late in the experiment, where initial sampling rate was between 10 and 20 Hz, following the improvements we saw more consistent sampling around 50 Hz. While this low sampling frequency is catastrophic in motion sensing tasks, the static environment and long sampling windows are largely unaffected by overall sampling rate as most information would be pure noise.

The water container used is a typical food-grade plastic container with an approximate capacity of 4 L and external dimensions of 30 cm L × 15 cm W × 10 cm H. The container is filled with exactly 3.0 L of normal drinking water. Distilled water would provide a stronger baseline difference, but we did not feel this was necessary for the experiment. We used normal table salt to create the various saline mixtures.

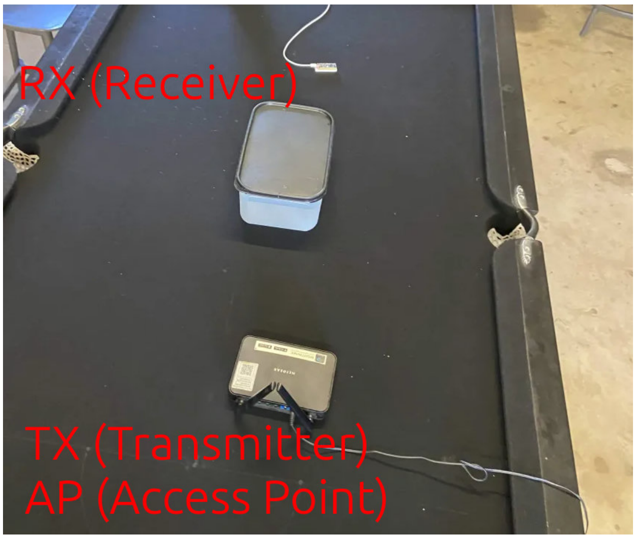

Figure 1: Recording setup for sessions 1-4. Each session relies on a different container position, which significantly affects the CSI geometry.

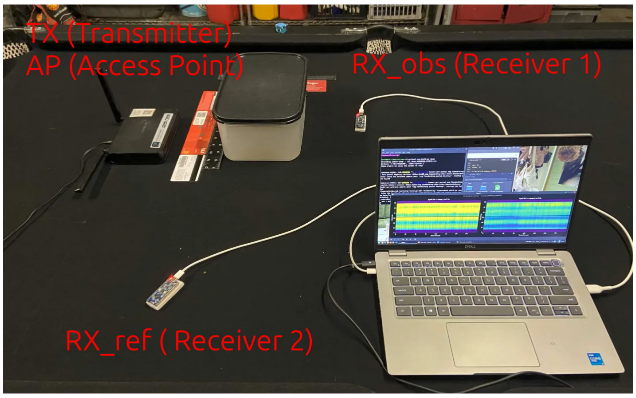

Figure 2: Recording setup for session 5. Key improvements include closer placement, repeatable container positioning jig, differential sensing control signal, and better Fresnel zone alignment for stronger signal.

The environment of the setup is an electromagnetically isolated shed, with no other active 2.4 GHz devices present and no moving objects during recording. This environment is better isolated to physical interference like human traffic, but is more susceptible

Recording	Concentration	Additive (cups)	Total approx. (grams)
1	0%	N/A	N/A
2	1%	1/8 cup	~30 g
3	3.5%	1/4 cup	~105 g
4	7%	3/8 cup	~210 g
5	15%	3/4 cup	~450 g

Table 1: Salt concentration per recording within a session.

to temporal drift as thermal expansion of the shed, wind, and air temperature cannot be appropriately controlled. The shed is metallic, generating constant and notable reflectance within propagation paths, and is fairly large, which distributes multipath signal as a background term in general recordings, but does not eliminate it. The low-frequency multipath effects theoretically allow us to isolate our high-frequency features more easily.

Each experiment progressed through salt concentrations by progressively adding salt in approximate volume increments, as appropriate scales were not available. The baseline concentration, at 0%, is placed within the propagation path between the two devices. When the recording is started, we ensure there is no avoidable interference activity within 30 meters of the devices, such as human activity or other movement. After 5 to 10 minutes, the recording is stopped, an incremental volume of salt is added to the solution, and the recording is restarted. This repeats until 15% concentration, where the experiment is complete for that session.

Data collection was distributed across five sessions and two days to isolate confounding factors and determine whether this sensing method is possible on ESP32 hardware. This can be understood as an iterative process to refine the result, and evaluate the performance of classification models on data with a low signal-to-noise ratio. Sessions 1-4 use the recording setup in 1, while session 5 uses a new approach based on observations within the first 4 sessions, as seen in Figure 2.

3.2.1 Session 1—Baseline Collection

The first session provides a meaningful baseline to determine if the shifting permittivity is measurable and separable within CSI data. We use 1 recording per stage, and a constant container position between stages. This recording was captured around 1pm, representing ideal daytime conditions for this experiment, as the temperature fluctuation is minimal.

Stage	Samples
Water	12633
1% Salt	13621
3.5% Salt	13711
7% Salt	13673
15% Salt	12479
Total	**66117**

Table 2: Samples collected at each concentration for session 1.

3.2.2 Session 2—Environmental Variation

The second session was a considerably longer collection aimed to test environmental variability. From approximately 3pm to 8pm, consistent recordings were taken across 5 random container orientations per stage. Adjusting the position of the container strongly affects the weak multipath signal, and with enough samples should theoretically allow the model to generalise. As the shed is relatively exposed to ambient temperature, the transition from day to night would also introduce temporal drift, that for a cross-session test would further validate the generalisability claim. The baseline water stage was only tested under one position.

Stage	1	2	3	4	5
Water	15318	N/A	N/A	N/A	N/A
1% Salt	11431	11630	8346	9265	13578
3.5% Salt	13104	13460	10619	10179	11478
7% Salt	11247	10337	9439	13345	13661
15% Salt	12145	12528	12906	11105	13337
Total					**248458**

Table 3: Samples collected at each position and concentration for session 2.

3.2.3 Session 3—Cross-session Control

The third session was similar to session 1, but is conducted at night where the thermal expansion artefacts that may be present in daytime tests are suppressed. The same procedure is followed, where the container has a constant position through the five stages. This test represents the similar but not identical case for model generalisability, where the full process is learned in this exact configuration to ideally isolate the target signal in model training.

Stage	Samples
Water	13514
1% Salt	13957
3.5% Salt	13385
7% Salt	13460
15% Salt	10838
Total	**65154**

Table 4: Samples collected at each concentration for session 3.

3.2.4 Session 4—Temporal Fingerprinting

Session 4 was a control test between the original setup for sessions 1-3, and for the new differential sensing setup, to determine whether machine learning models could classify labels with no obvious or intentional signal. In the first case, we record 14 minutes of samples from a single ESP32 during the day, and in the second case, 37 minutes of samples are recorded for two ESP32s at night. To label this data, we split the timeseries uniformly and train various models on these labels within the session—if a model is able to classify them correctly, then a detectable temporal drift is present in the data.

This discrimination was tested on both classical machine learning and deep learning models—particularly, a Support Vector Machine and a Convolutional Neural Network were used. The demonstration is twofold that, when presented with low signal-to-noise ratio data, both methods are susceptible to 'shortcut learning' of undesirable features.

Setup	Time (min)	Samples
Day (1 ESP32)	14.2	13937
Night (2 ESP32s)	37.8	213007
Total	52	**226944**

Table 5: Day and night control samples across both testing setups.

3.2.5 Session 5—Differential Sensing

The fifth and final session aims to consolidate the confounding signals identified in the previous four tests, with a design that—while imperfect in terms of classification—is a more fair representation of the system's water salinity detection properties. In this test, two passes of the experiment were recorded in immediate succession, with a small fixture to align the container between runs with ± 1 mm precision and mitigate geometric drift between sessions.

A second ESP32 is also deployed to collect a control signal and mitigate session-specific temporal drift from training data, to then improve cross-session generalisation for the intended signal. The second session is intended to inherit the thermal profile at the end of the first session, theoretically producing a different thermal profile that will not contaminate the data.

For this test, we also modified the data collection pipeline to include the complete complex I/Q data, rather than just amplitude, which allows us to perform the complex division needed for differential sensing. The modified program also better manages the serial stream, and has a higher effective throughput of about 50 Hz per-device.

Stage	Train	Test
Water	108469	56224
1% Salt	50493	78787
3.5% Salt	75685	72903
7% Salt	66242	80674
15% Salt	82997	54825
Total	**383866**	**343433**

Table 6: Sample counts for train and test procedures within session 5, accounting for two ESP32 devices.

3.3 Data Processing and Model Fitting

ESP32 CSI data in this experiment is very low signal-to-noise, therefore extensive filtering techniques are required to make the data usable. Between the first and second recording sessions, we made significant revisions to the collection and classification pipelines following analysis of prior results. Both methods trim the first and last 30 seconds of recording data to exclude setup and teardown transients. For clarity, sessions 1-4 utilise the first formulation, while session 5 utilises the second formulation.

3.3.1 First Formulation

Raw CSI frames are time aligned by linear interpolation to a fixed cadence of 10 Hz. The sampling rate of this version was very low (10-20 Hz) and the rhythm of sampling frames may be detectable as a session fingerprint, through factors like temperature of both the device and the recording computer. We aimed to smooth the temporal cadence in order to mitigate this.

During this experiment, the temporal drift of the devices had not yet been identified, and the correct signal processing technique was unclear. In preprocessing, we would divide the sessions into fixed-width windows (20, 50, 100, 200) to evaluate the effect of window width on model accuracy. Within these windows, we tried several filtering techniques, including log scaling, bandpass, median, minmax and Savitzky-Golay filters, and both temporal and cross-subcarrier normalisation.

Fitting a model on the data is more exploratory in these experiments, as we have not yet identified the correct model for the constraints at this point. We created pipelines for both an SVM and CNN models to evaluate classification directly to the labels, and tested several configurations of data from the first three sessions, including the contrasting result that led to the session fingerprinting and shortcut learning observations, being train-test splitting within the same *session*, and cross-session splitting on different *positions*.

3.3.2 Second Formulation

Raw CSI frames are time-aligned using nearest-neighbour matching with a 50 ms tolerance, to compensate for inconsistent sampling frequencies across two devices.

To address the per-frame noise floor imposed by the ESP32 integer I/Q quantisation, features were extracted from sliding windows of $N = 64$ consecutive frames ($\sim$2.5 s at approximately 25 frames/s). From each window, the following 216-dimensional feature vector was computed:

- Mean amplitude ratio $|\Delta H_k|$ per active subcarrier (54 values)
- Standard deviation of amplitude ratio over the window (54 values)
- Mean sanitised differential phase per subcarrier (54 values)
- Standard deviation of differential phase over the window (54 values)

The standard deviation values specifically carry physical information, as water salinity affects both the phase and amplitude of the signal, which should be detectable in the temporal signature of the noise. Each feature vector was L2-normalised before classification.

This feature space relies heavily on the differential phase component to bypass the hardware limitations of the ESP32. As established in Equation 6, raw phase is severely distorted by time-varying CFO and STO parameters ($mk + b$). As both the reference and observation devices capture packets simultaneously on a shared receiver architecture, they also inherit near-identical hardware clock trajectories and thermal drift. By using complex division to compute the differential channel response $\Delta H_k = H_{\mathrm{obs},k}/H_{\mathrm{ref},k}$, these contaminated profiles are subtracted in the exponent.

$$\hat{\phi}_{\mathrm{obs},k} - \hat{\phi}_{\mathrm{ref},k} = (\phi_{\mathrm{obs},k} + mk + b) - (\phi_{\mathrm{ref},k} + mk + b) \quad (10)$$

The shared linear slope m and intercept b cancel out, isolating the true physical phase difference caused by dielectric properties of the saline solution. With the feature vectors stabilised against this drift, we can then pass them to predictive models.

An SVM with an RBF kernel was trained for the five-class problem, with hyperparameters C and γ selected by five-fold cross validation on the training session and `class_weight=balanced` to account for unequal window counts between conditions. An SVR with the same kernel was trained on an ordinal encoding of the five concentrations (0, 1, 3.5, 7, 15 g/100 mL) to produce a continuous concentration estimate. The regression model specifically is selected to more appropriately model the monotonic change with salinity, forcing the model to learn a continuous feature rather than a discrete classification.

4 Results

4.1 Phase 1—Ablation Study

In the initial experimentation phase, raw CSI amplitude and phase features from the observed link were used without differential processing or cross-session testing. A classifier was trained and evaluated using the standard within-session protocol, where training data and testing data were drawn from non-overlapping time windows of the same recording session. Theoretically, this split should not contaminate the data, however this is the first sign of the 'session fingerprint' we discuss in this report.

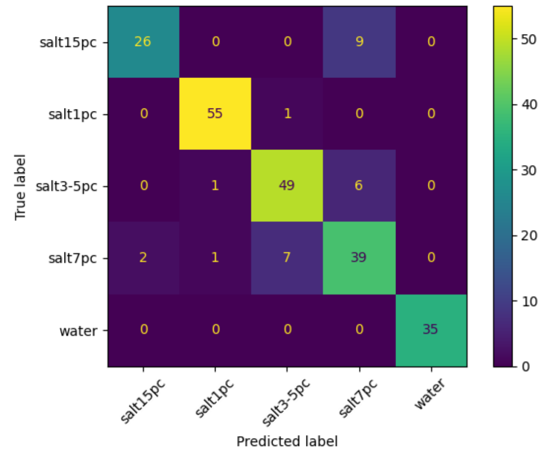

Figure 3: Confusion matrix for Session 1 recording data. Split random non-overlapping windows. Best reported accuracy: 0.833

This accuracy is suspiciously high for the problem, the next logical step is to verify generalisation across several different positions, at different times of day. Applying the same random non-overlapping split yields 0.995 accuracy, which is even higher than the single-session test.

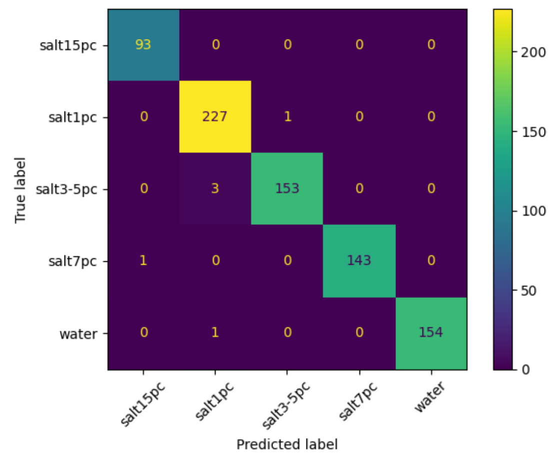

Figure 4: Confusion matrix for multi-session combined test.

If we interpret this result at face value, then this is an absolute success. Critically, however, we must consider what this test is actually classifying. Moving the container to a distinct position for each recording has a far greater effect than the permittivity change or the temporal drift, and because CNNs can effectively classify sub-features of a superclass (by design), the model is able to reliably classify the containers by positions that are in the training set. Furthermore, because the effect of temporal drift is far weaker than that of geometry, the test set shares both the same geometric and temporal fingerprint as the training set, providing the model with information that would not be available in a genuine cross-session deployment. If we modify this test and remove a small subset of the position recordings (2 per level), we observe total model collapse:

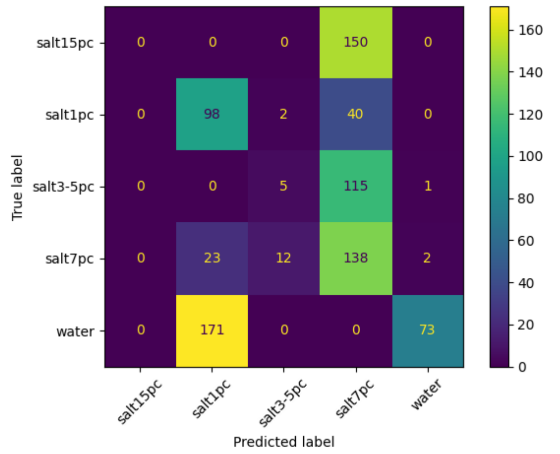

Figure 5: Model collapse on left-out cross-session test set. Best accuracy with a CNN was 0.284, which is barely higher than chance for a 5-choice problem. During training, the model did not converge and finished 50 epochs with 0.136 accuracy.

With this information, we now understand the dominant effect of geometry in CSI data, so in order to isolate our salinity signal, we must also strictly control for geometry. In an experiment similar to session 1, we observe within-session accuracy of 0.911 (Figure 7), consistent with the Session 1 result. Training on Session 1 and evaluating on Session 3, under the same physical setup at a different time of day and slightly different container position, yields exactly chance accuracy of 0.2 (Figure 6). The model has learned nothing transferable about the physical signal. Further, the cross-session test consistently guesses the same label in the test set, which suggests a more specific failure than model collapse where instead of guessing randomly, all samples in the test session appear most similar to 'salt1pc' of the training session, which implies the model has learned something other than the target signal.

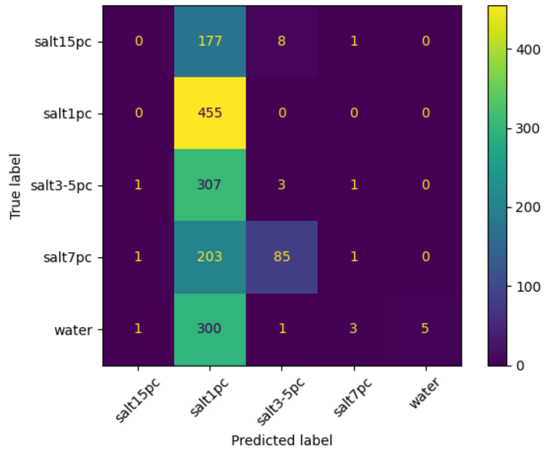

Figure 6: Train test split between session 1 and 3. Same experiment, slightly different position and time of day. Accuracy is exactly chance at 0.2, confusion matrix suggests fingerprint of 'salt1pc' label is most similar to daytime session fingerprint.

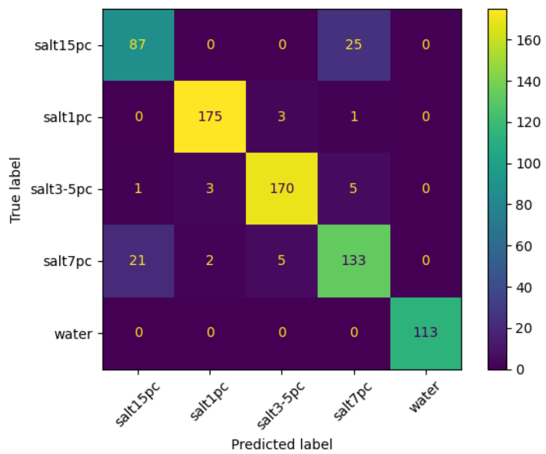

Figure 7: Session 3 test, within-session split. Reported accuracy: 0.911

To formulate a new experiment that best controls for position, we must also understand what proportion of within-session accuracy is attributable to temporal drift rather than the target signal. To measure this, we labeled a baseline recording with no variation in salt concentration, splitting it into uniform time segments and training classifiers to discriminate between them. If a model can classify arbitrary time segments in data with no physical label variance, the discriminating feature is then some form of session fingerprint. Both a CNN and SVM were evaluated on the same data to determine whether this susceptibility is model-specific.

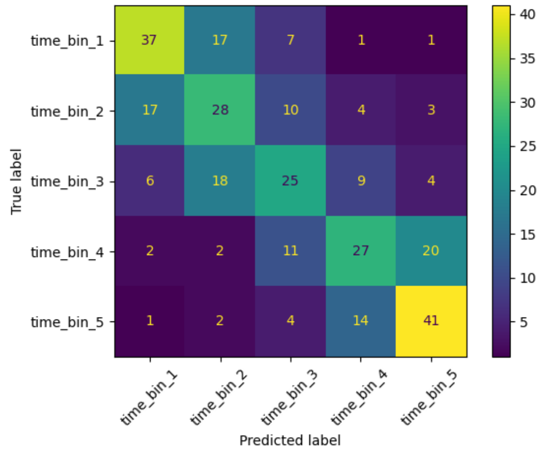

Figure 8: Model fitting to temporal drift, setup 1, CNN classifier. Best accuracy: 0.55

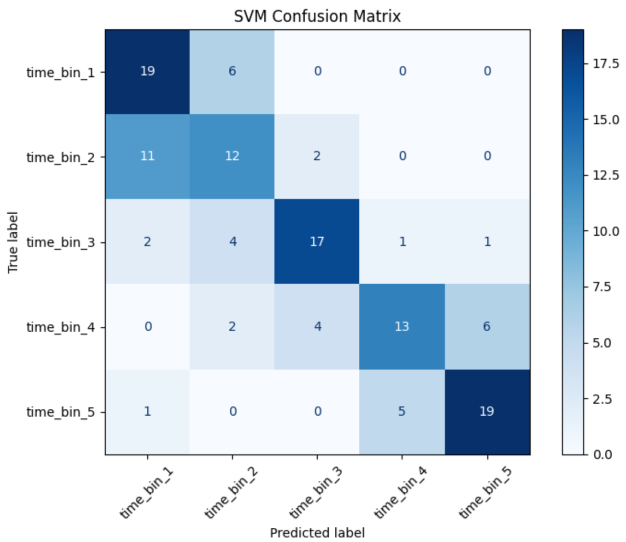

Figure 9: Model fitting to temporal drift, setup 1, SVM classifier. Best accuracy: 0.64

Both models substantially exceeded the chance rate of 0.20 for a five class problem—the CNN achieves 0.55 and the SVM achieves 0.64. Neither result is high in absolute terms, but both are significantly above chance on a task where no valid signal exists. When using within-session test splits, there is no meaningful way to distinguish whether a trained model is classifying on the target signal or the session fingerprint, or where accuracy is otherwise inflated by some combination of both. A cross-session test must be used to validate such claims.

This baseline test was repeated under the new differential sensing setup with two ESP32 devices, confirming that temporal drift is present and classifiable in both the observed and reference links independently. This result validates the drift as a physical property of CSI sensing, and directly motivates the use of differential sensing so the shared fingerprint may be suppressed in the differential channel.

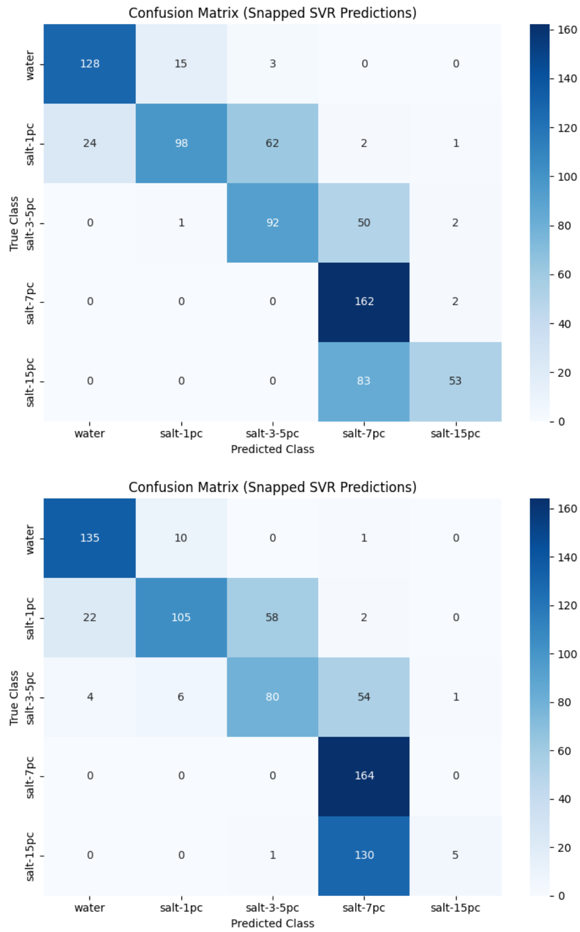

Figure 10: Model fitting to temporal drift in new experiment, for both observed and reference devices. Category snapping of SVR may present classification artefacts in this visualisation.

4.2 Phase 2—Differential Sensing and Cross-Session Fitting

Having established that within-session accuracy is not a valid performance measure, Phase 2 applied the differential CSI pipeline and evaluated exclusively in the cross-session space. For the observed stream, the complex channel is divided by the simultaneously unobstructed reference stream. This suppresses shared session fingerprint as hardware drift effects, in principle, affect both devices similarly. The model then must generalise on whatever signal remains within the obstructed path, which corresponds more strongly to dielectric response of the container contents than previous experiments.

4.2.1 Cross-Session Ablation

To support this experiment, the individual signals are also validated independently in a machine learning model (SVR) under both within-session and cross-session evaluation. We can readily observe that the hypothesis of within-session shortcut learning, and cross-session model collapse is validated for the control signal, suggesting a separable session fingerprint within each recording. The control signal demonstrates different behaviour to previous cross-session tests, where the observed salinity gradient is exactly inverted across sessions, producing a less-than-chance MAE of 8.72% in the regression model. This inversion suggests outer Fresnel zone interference of the container on the control signal—if the reference link carries a weak salinity-dependent component, the session-specific phase of that component may reverse between runs as the thermal state of the hardware changes, producing a gradient that opposes rather than mirrors the training distribution. This is explored further in Section 4.2.4

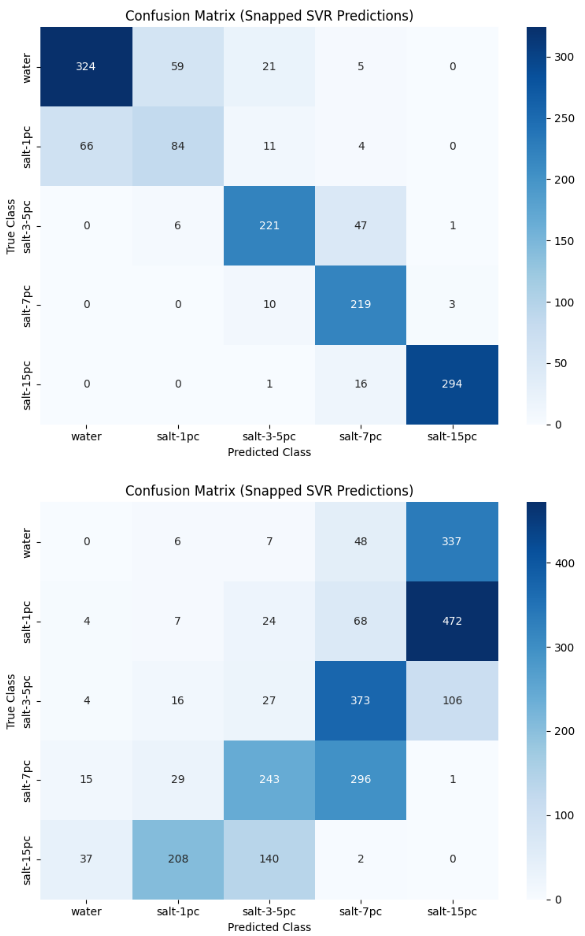

Figure 11: Within-session and cross-session confusion matrices for the control signal used in Phase 2. Within-session test reports 1.03% MAE, while cross-session test reports 8.72% MAE.

For the observed signal, the cross session test independently suggests that the target signal may be observed, as the model does not collapse on cross-session tests, however this alone is not sufficient to validate the claim.

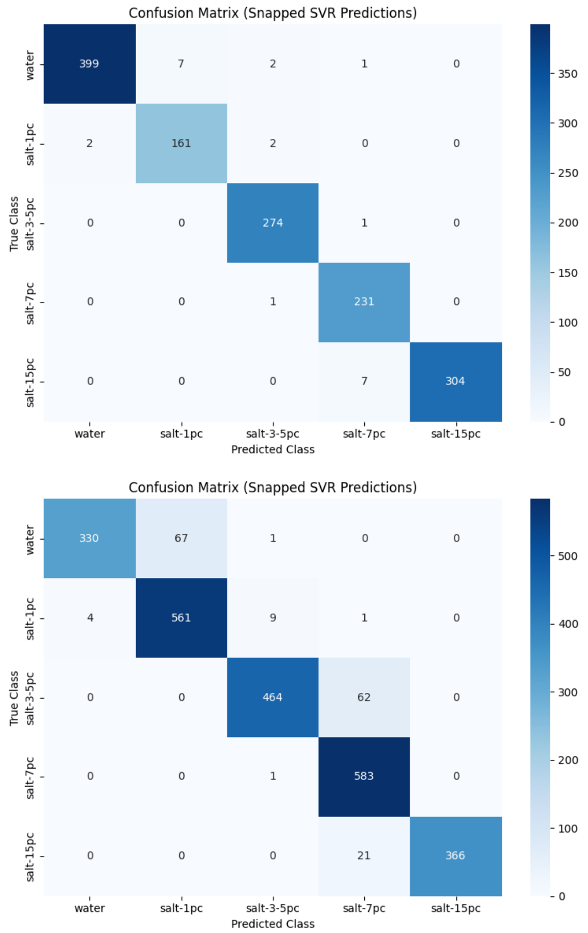

Figure 12: Within-session and cross-session confusion matrices for the observed signal used in Phase 2. Within session test reports 0.22% MAE, while cross-session test reports 0.94% MAE.

4.2.2 Classification Results

Applying the SVM classifier to the differential CSI features under cross-session evaluation yields a prediction accuracy of 0.61 and mean absolute error of 1.82%. This represents a substantial recovery relative to the cross-session performance in Phase 1.

Class	Precision	Recall	f1
Water	0.87	0.99	0.93
1% Salt	0.97	0.16	0.27
3.5% Salt	0.50	0.98	0.66
7% Salt	0.56	0.84	0.68
15% Salt	0.81	0.03	0.06
Weighted Avg.	0.73	0.61	0.52
Accuracy			**0.61**

Table 7: Cross-session classification performance metrics for the Support Vector Classifier (SVC) with differential sensing.

These results demonstrate the physical reality of skin depth at 2.4 GHz in salt water—as pure water already has a very shallow skin depth at this frequency, the increased conductivity of the medium has a diminishing effect on the signal as salt concentration increases. Pure water and 3.5% concentration are highly separable, but the model almost never guesses 15% because the physical difference between 7% and 15% is vanishingly small, relative to 0% and 3.5%. The water recall of 0.99 reflects the model heavily favouring the water class. It correctly identifies nearly all water samples, but at the cost of absorbing borderline 1% samples as false negatives, contributing to a recall of only 0.16 for that class. 1% concentrations in this case are largely misclassified as 3.5%, and 15% concentrations are misclassified as 7%, highlighting stronger feature-space separability of these concentrations. These misclassifications highlight that hard categorical boundaries may be fundamentally incompatible with modeling continuous dielectric transitions.

4.2.3 Regression Results

The SVR trained on an ordinal encoding of the five concentrations yields a mean absolute error of 1.35% on the cross-session test set. After rounding continuous predictions to the nearest labeled concentration, the resulting classification accuracy is 0.73. We use this result for direct comparison with the SVC measurement, although MAE is the more physically meaningful metric. The regression formulation also appears more physically appropriate than classification for this task, and for classes where categorical classification struggled to separate adjacent classes, regression more reliably separates the gradient of salinity.

Class	Precision	Recall	f1
Water	0.83	0.09	0.16
1% Salt	0.56	0.81	0.66
3.5% Salt	0.81	0.94	0.87
7% Salt	0.77	0.97	0.86
15% Salt	1.00	0.65	0.79
Weighted Avg.	0.78	0.73	0.69
Accuracy			**0.73**

Table 8: Cross-session regression and class-snapping classification performance metrics for the Support Vector Regressor (SVR) with differential sensing.

While these results initially suggest a localised degradation in modeling the low-concentration complex permittivity gradient—most notably illustrated by the inversion of the pure water and 1% salt recall relative to the classifier—this is most likely an artefact of the post-processing category snapping function rather than a failure of model fitting. Numerically, the interval between 0.0% and 1.0% is highly narrow, meaning minor baseline shifts will disproportionately affect this range. If the model predicts 0.51% salt on a pure water sample, it is classified as 1% salt, while at higher concentrations higher absolute error is tolerated.

Furthermore, the continuous loss function successfully resolves the non-linear attenuation floor at the upper bound, lifting recall of the 15% solution from 0.03 to 0.65. The classification performance of continuous regression is significantly improved over direct classification for this task, which supports the hypothesis that continuous feature-space tracking is essential for effective permittivity gradient sensing in dielectric materials.

4.2.4 Signal Erosion in Differential Sensing

When evaluating model performance on cross session data using the H_{obs} receiver data in isolation, we observe a significantly improved accuracy at 0.93 and mean absolute error of 0.94%.

Class	Precision	Recall	f1
Water	0.99	0.83	0.90
1% Salt	0.89	0.98	0.93
3.5% Salt	0.98	0.88	0.93
7% Salt	0.87	1.00	0.93
15% Salt	1.00	0.95	0.97
Weighted Avg.	0.94	0.93	0.93
Accuracy			**0.93**

Table 9: Cross-session SVR performance on raw observed channel H_{obs} features, without differential processing.

While operating without controlled differential sensing introduces the theoretical risk of recording artefact contamination, such as hardware gain steps or thermal drift, the overwhelming cross-session resilience shown in Table 9 strongly indicates the model is tracking a coherent physical phenomenon. Confounding hardware and environmental artefacts typically manifest as chaotic, session-specific baselines that cause cross-session model collapse. In this case, SVR correctly maps five distinct concentrations across independent data collection sessions, which demonstrates that these raw features retain a stable, monotonic physical signature modulated by complex permittivity of the water.

Most notably, evaluating H_{obs} in isolation completely resolves the low-concentration resolution failure observed in the differential SVR framework. In the differential experiment, recall collapsed to 0.09 due to baseline compression. The raw channel receives a precision of 0.99 and recall of 0.83 for water, and a recall of 0.98 for 1% salt. This confirms that the sharp change in ionic conductivity produces a high-amplitude feature that is easily tracked when the dynamic range of the signal is uninhibited.

The disparity reveals a critical geometric constraint in the differential sensing layout. As illustrated in the experimental topology of Figure 2, the outer Fresnel zones of the propagation path of the reference receiver H_{ref} has inadvertent collision with the saltwater volume. Although the direct line-of-sight signal between the transmitter and the reference antenna remains dominant, the proximity of the conductive saline medium induces a non-zero multipath scattering and diffraction effect on the reference CSI data.

Consequently, when the ratiometric complex division $(\frac{H_{\text{obs}}}{H_{\text{ref}}})$ is applied to eliminate common-mode hardware drift, it mathematically subtracts a meaningful fraction of the target salinity signal. This signal erosion compresses the feature space significantly, driving the delicate subcarrier variations at the extreme boundaries beneath the resolvable quantisation threshold of the ESP32. While the differential architecture remains conceptually superior for long-term drift rejection, the raw single-device channel delivers better immediate performance within the controlled conditions of Session 5 by preserving the full dynamic range of the dielectric interaction. Without differential processing however, this performance is highly contingent on session-to-session stability. The Phase 1 results demonstrate that raw CSI features collapse under greater thermal or geometric variability. Differential sensing remains the more principled architecture for deployment conditions where such stability cannot be guaranteed.

5 Discussion

The incremental results of this investigation reveal critical insights into the coupled constraints of microwave physics, commodity Wi-Fi devices, and statistical learning. Phase 1 experiments yielded data that is heavily confounded by environment variables, and their systemic failure modes provide a necessary diagnostic framework for understanding the physical and practical limits of commodity CSI Wi-Fi sensing.

5.1 Domain Constraints

Dielectric material sensing is unique from other common CSI sensing activities, such as Human Activity Recognition (HAR) [2], because of the static temporal signature required for identifying material environments. In a sense, HAR requires a frame of reference for previous samples within a sequence, and relies on dynamic and predictable signal perturbations over time to identify a pattern. Material sensing, conversely, can only derive identity from contrasting frames in a separate sequence. As there is no clear perturbation induced over time by dielectric material in the general case, the signature of a material signal is deeply embedded within the multipath structure of the environment itself. Separating this structure requires strong separation from both the high-frequency noise floor, and low-frequency temporal drift, which is a very fragile problem of signal processing. HAR models are less sensitive to these effects.

5.2 Session Fingerprints and Shortcut Learning

The session fingerprint is a pervasive confound that contaminates a significant block of commodity Wi-Fi CSI data, with effects that are rarely accounted for in the literature [20–23]. Session fingerprints are defined as a low-frequency background signal that varies with time—as most tasks cannot be recorded in a controlled manner simultaneously, this signal manifests uniquely within every recording session. The session fingerprint is comprised, generally, of three components:

1. Thermal trajectory of the hardware that affects the transmitter and receiver oscillators, slightly altering the characteristics of the true signal and noise profile over time.
2. Thermal trajectory of the environment subtly affects its geometry, leading to subtle perturbation of a multipath signal that introduces significant error. As geometry is identified to be a primary signal contributor in CSI data, the thermal properties of reflective surfaces like metal walls has a non-zero effect on CSI data.
3. Hardware gain states such as Automatic Gain Control (AGC) and other low-frequency interference can introduce discrete, session-specific boundaries within data.

In terms of signal, these factors trivially affect the intended operation of the device. For CSI sensing, highly expressive machine learning models like Support Vector Machines and Convolutional Neural Networks are hypersensitive to these statistical discrepancies. When a model is evaluated using a train-test split, where samples are drawn from the same recording block, the model exploits these session-specific boundaries rather than learning the physical properties of the target.

Our empirical results demonstrate this artefact. For within-session splits, the model consistently achieves prediction accuracy of 0.99 or higher, even with variant positions. Initially, a single-session baseline accuracy of 0.9 was assumed to be a plausible reflection of true dielectric sensing, but when multiple spatial positions were aggregated into the dataset, the within-session accuracy anomalously rose to over 0.995. Subsequent analysis revealed that environmental geometry yields an order-of-magnitude stronger CSI signature than any contiguous complex permittivity shifts. Rather than generalising the underlying relationship between salinity and signal attenuation, these expressive classifiers merely memorised the discrete geometric and thermal fingerprints unique to each session state. This was proven with a held-out test set, where the model predicted at just chance accuracy of 0.2.

In much of the contemporary CSI literature, these metrics are reported prematurely and without methodological scrutiny. This artificial inflation of accuracy, followed by catastrophic model collapse when evaluated on a held-out cross-session validation set, is a clear manifestation of "shortcut learning" [15]. On commodity architectures like the ESP32, the session fingerprint is exceedingly difficult to suppress, which induces deceptive results with no physical basis. Stated plainly, literature on commodity Wi-Fi sensing often accidentally claims certain physical phenomena may be differentiable in CSI without any supporting evidence. At the very least, cross-session evaluation should be used to validate any claims [17].

This vulnerability can be further demonstrated by our baseline control tests, where classical machine learning algorithms could achieve an accuracy of 0.7 when classifying arbitrary temporal segments—in a completely static recording—with no target signal whatsoever. If the complexity of the target signature is smaller than the mathematical variance of the session fingerprint, statistical models will invariably select the path of least resistance. While cross-session validation eliminates this false positive, techniques like differential sensing can meaningfully increase the signal-to-noise ratio and improve real classification.

5.3 Limitations

Despite the cross-session resilience achieved in our final models, several unresolved architectural constraints limit the immediate interpretability of these results. First, the precise electromagnetic interaction mechanism is driven by surface reflections and wide-angle edge diffractions from the liquid container, rather than the direct through-path transmission originally designed. A misidentification of the skin depth of water at 2.4 GHz prefaced the design of this experiment, and the physical interpretation of these results is justified retroactively. The reliance on multipath scattering limits our ability to extract quantitative physical properties that would be tractable in a direct transmission or wave guiding experiment.

Second, the spatial overlap between the Fresnel zone of the reference receiver link (H_{ref}) and the target medium remains an unresolved oversight and geometric artefact that actively introduces signal erosion.

Finally, our successful cross-session evaluation in Session 5 relies on a sequential, back-to-back data collection pair rather than a broad, multi-day longitudinal study. Consequently, the results of this research are not necessarily conclusive on the material sensing characteristics of ESP32 CSI data, however we have laid the groundwork for more effective experimentation in the future.

6 Future Work

The limitations of this work directly motivate three avenues of future work. First, we should aim to use more precise instrumentation in a future experiment, to limit the relative error introduced by confounding variables. For example, using calibrated scales to measure salt and water quantities eliminates the imprecise measurement used in this experiment as a failure mode, likely producing a clearer signal. Isolated laboratory conditions would also reduce the noise floor within the experiment, and although the high-noise conditions are more reflective of a realistic deployment, improved conditions and instrumentation allow more detailed analysis of results.

Second, the reference device placement used in Session 5, as we identified, incidentally overlaps with the container and introduces signal erosion. We observe this in Section 4.2.4, where the accuracy loss in the differential feature space is likely a direct consequence of signal erosion, because the target signal is present in our reference. Properly isolating the reference link from the target with a metal reflector or different positioning would provide a more stable baseline signal, restore the full dynamic range of the dielectric signature, and improve the performance of machine learning models on the differential feature space.

Third, the cross-session evaluation should be extended to a larger number of independent sessions, ideally across different days and ambient temperature ranges, to characterise the residual session-to-session variability of the differential channel under realistic deployment conditions. To explore further generalisation, we can also investigate Domain Adversarial Neural Networks to extend the utility of existing feature-space embeddings.

7 Conclusion

This paper has demonstrated that NaCl concentration in a water sample can be resolved from differential ESP32 CSI data under cross-session evaluation, achieving 0.61 classification accuracy and 1.35% MAE with SVR after differential preprocessing. The regression formulation was shown to be more physically appropriate than classification for this task, recovering the monotonic salinity gradient that hard categorical boundaries suppress.

More significantly, we demonstrated a systematic methodological confound in CSI sensing literature. Within-session evaluation produces accuracy figures—up to 0.995 in our experiments—that collapse to chance when evaluated on held-out sessions. This confound persists for both classical machine learning and deep learning, where any sufficiently expressive model can exploit a session fingerprint when evaluated within-session. Cross-session evaluation is a necessary condition for any valid CSI sensing claim, and results reported without it cannot be taken as evidence of physical sensing capability.

The signal erosion analysis revealed an additional constraint. Differential sensing architectures require geometric separation between the reference link and the sensing target. Fresnel zone overlap of the reference receiver with the target volume partially cancels the salinity signal in the differential channel, compressing the feature space at low concentrations. Future designs should treat reference link placement as a primary geometric constraint, alongside transmitter and receiver alignment.

References

[1] X. Wang, L. Gao, S. Mao, and S. Pandey, "DeepFi: Deep learning for indoor fingerprinting using channel state information," in *2015 IEEE Wireless Communications and Networking Conference (WCNC)*, pp. 1666 – 1671, IEEE, 2015.

[2] J. Niu, X. He, B. Fang, G. Han, X. Wang, and J. He, "CSI-F: A Human Motion Recognition Method Based on Channel-State-Information Signal Feature Fusion," *Sensors*, vol. 24, p. 862, Jan. 2024.

[3] H. Song, B. Wei, Q. Yu, X. Xiao, and T. Kikkawa, "WiEps: Measurement of Dielectric Property With Commodity WiFi Device—An Application to Ethanol/Water Mixture," *IEEE Internet of Things Journal*, vol. 7, pp. 11667–11677, Dec. 2020.

[4] K. Nörtemann, J. Hilland, and U. Kaatze, "Dielectric Properties of Aqueous NaCl Solutions at Microwave Frequencies," *The Journal of Physical Chemistry A*, vol. 101, pp. 6864–6869, Sept. 1997.

[5] S. M. Hernandez and E. Bulut, "WiFi Sensing on the Edge: Signal Processing Techniques and Challenges for Real-World Systems," *IEEE Communications Surveys & Tutorials*, vol. 25, no. 1, pp. 46–76, 2023.

[6] D. M. Pozar, *Microwave engineering.* Hoboken, NJ: John Wiley & Sons, Inc, fourth edition ed., 2012.

[7] J. Baker-Jarvis and S. Kim, "The Interaction of Radio-Frequency Fields With Dielectric Materials at Macroscopic to Mesoscopic Scales," *Journal of Research of the National Institute of Standards and Technology*, vol. 117, pp. 1–60, 2012.

[8] F. E. Harris and C. T. O'Konski, "Dielectric Properties of Aqueous Ionic Solutions at Microwave Frequencies," *The Journal of Physical Chemistry*, vol. 61, pp. 310–319, Mar. 1957.

[9] K. Omeke, A. Abohmra, M. Imran, Q. Abbasi, and L. Zhang, "Characterization of RF signals in Different Types of Water," in *Antennas and Propagation Conference 2019 (APC-2019)*, (Birmingham, UK), pp. 6 pp.–6 pp., Institution of Engineering and Technology, 2019.

[10] A. Stogryn, "Equations for Calculating the Dielectric Constant of Saline Water (Correspondence)," *IEEE Transactions on Microwave Theory and Techniques*, vol. 19, pp. 733–736, Aug. 1971.

[11] R. Buchner, G. T. Hefter, and P. M. May, "Dielectric Relaxation of Aqueous NaCl Solutions," *The Journal of Physical Chemistry A*, vol. 103, pp. 1–9, Jan. 1999.

[12] IEEE, "IEEE Standard for Information technology– Local and metropolitan area networks– Specific requirements– Part 11: Wireless LAN Medium Access Control (MAC)and Physical Layer (PHY) Specifications Amendment 5: Enhancements for Higher Throughput," *IEEE Std 802.11n-2009 (Amendment to IEEE Std 802.11-2007 as amended by IEEE Std 802.11k-2008, IEEE Std 802.11r-2008, IEEE Std 802.11y-2008, and IEEE Std 802.11w-2009)*, pp. 1–565, 2009.

[13] X. Wang, L. Gao, and S. Mao, "CSI Phase Fingerprinting for Indoor Localization With a Deep Learning Approach," *IEEE Internet of Things Journal*, vol. 3, pp. 1113–1123, Dec. 2016.

[14] I. Guarino, D. Carra, M. Cominelli, F. Gringoli, and R. Lo Cigno, "A survey on CSI-based Wi-Fi sensing datasets and models with a focus on reproducibility," *Computer Communications*, vol. 249, p. 108431, Mar. 2026.

[15] R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, and F. A. Wichmann, "Shortcut Learning in Deep Neural Networks," *Nature Machine Intelligence*, vol. 2, pp. 665–673, Nov. 2020. arXiv:2004.07780 [cs.CV].

[16] V. V. Ratnam, H. Chen, H.-H. Chang, A. Sehgal, and J. Zhang, "Optimal Preprocessing of WiFi CSI for Sensing Applications," *IEEE Transactions on Wireless Communications*, vol. 23, pp. 10820–10833, Sept. 2024.

[17] A. Brunello, A. Montanari, R. Montoliu, A. Moreira, N. Saccomanno, E. Sansano-Sansano, and J. Torres-Sospedra, "Time matters: Empirical insights into the limits and challenges of temporal generalization in CSI-based Wi-Fi sensing," *Internet of Things*, vol. 32, p. 101634, July 2025.

[18] C. Cortes and V. Vapnik, "Support-vector networks," *Machine Learning*, vol. 20, pp. 273–297, Sept. 1995.

[19] H. Drucker, C. J. C. Burges, L. Kaufman, A. Smola, and V. Vapnik, "Support Vector Regression Machines," in *Advances in Neural Information Processing Systems* (M. C. Mozer, M. Jordan, and T. Petsche, eds.), vol. 9, MIT Press, 1996.

[20] S. Mosleh, J. B. Coder, C. G. Scully, K. Forsyth, and M. O. Al Kalaa, "Monitoring Respiratory Motion With Wi-Fi CSI: Characterizing Performance and the BreatheSmart Algorithm," *IEEE access: practical innovations, open solutions*, vol. 10, pp. 131932–131951, 2022.

[21] H. S. Maghdid, S. R. H. Salah, A. T. Hawre, H. M. Bayram, A. T. Sabir, K. N. Kaka, S. G. Taher, L. S. Abdulrahman, A. K. Al-Talabani, S. M. Asaad, and A. Asaad, "A Novel Poisoned Water Detection Method Using Smartphone Embedded Wi-Fi Technology and Machine Learning Algorithms," Feb. 2023. arXiv:2302.07153 [eess.SP].

[22] J. Li, *Smart WiFi Sensing for Network and Environmental Awareness.* PhD thesis, UNSW Sydney, 2025.

[23] A. M. Alfadhel and A. A. Hilli, "Water Quality Evaluation Through Channel State Information in WiFi Networks," in *2021 International Conference on Engineering and Emerging Technologies (ICEET)*, (Istanbul, Turkey), pp. 1–5, IEEE, Oct. 2021.